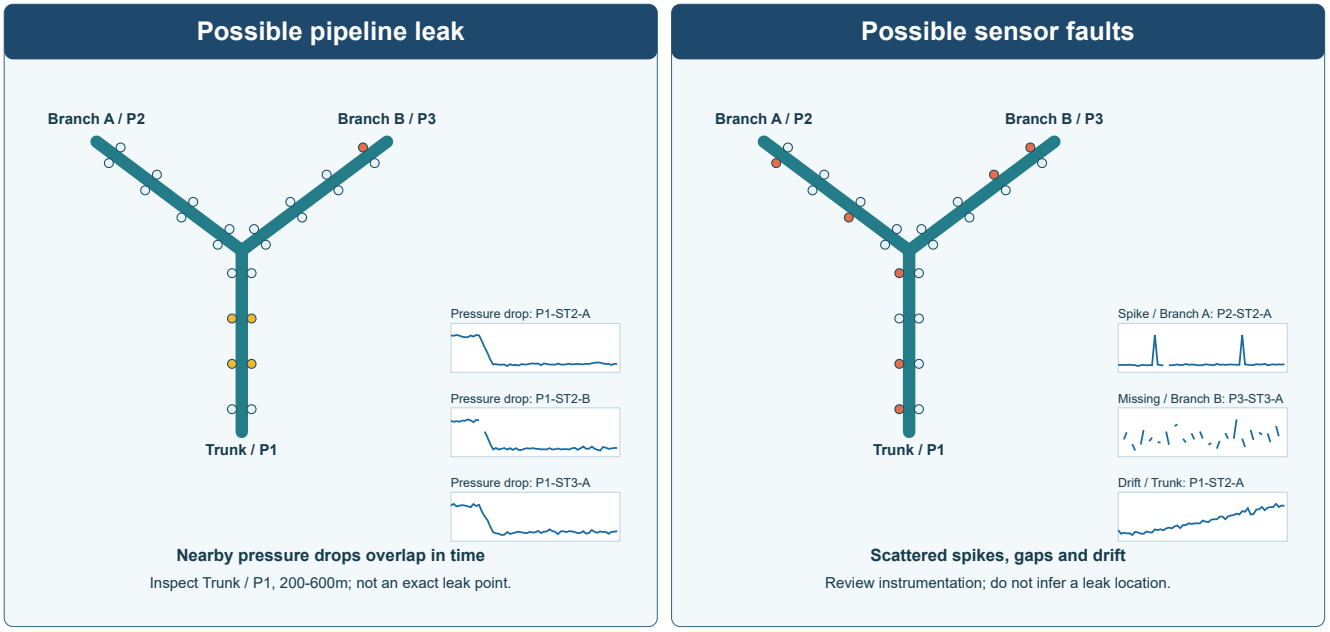
Subsea Integrity Copilot - Review Brief

Customer and problem. NorthSea Integrity Operations is a fictional offshore gas operator that needs to monitor a long subsea pipeline network. A leak could release gas into the environment and interrupt operations, so warning signs must be investigated promptly. To support this, many sensors distributed along the pipelines continuously stream pressure readings, and existing ML algorithms flag unusual changes.

The difficulty is that the monitoring equipment can also malfunction. A faulty sensor may produce sudden spikes, stop reporting measurements or gradually drift away from its expected readings even when the pipeline is healthy. Engineers therefore need to assess the quality of the measurements while investigating a possible leak. Rather than judging each alert in isolation, they must compare the timing and location of anomalies across several sensors and relate those patterns to the pipeline layout.

The diagram illustrates this reasoning. On the left, several nearby sensors show related pressure drops within the same monitoring window. This spatial and temporal agreement supports a possible leak investigation and helps narrow the section that may need inspection. By contrast, the scattered spikes, missing readings and drift on the right suggest individual sensor faults rather than one localized pipeline event. Neither pattern is conclusive: operating changes can affect several sensors, and shared communication or equipment failures can also produce related anomalies.

Today, this investigation depends on a few experienced engineers who combine sensor histories, pipeline layout and operational knowledge. Their expertise is difficult to obtain, translate into software and pass to newcomers. At scale, manual analysis takes time and important warnings may be overlooked among sensor-related alerts. The proposed solution makes documented expert knowledge reusable in agent-assisted investigation: distinguish sensor faults from possible leaks, narrow down where to inspect and explain the evidence to less-experienced engineers. Initial numerical warnings do not wait for an AI report. It builds on existing detectors; humans retain safety authority.



Existing detector alerts → Agent-assisted investigation → Human review

Patterns are evidence, not proof: check operating changes and shared failures.

Demo assumptions: 24 pipe-mounted pressure instruments, 12 paired stations, three connected 800m segments.

Same pressure-pack-v3 geometry as console; seed 2026, [80,140)s, pressure in bar; independent trace y-scales; synthetic.

Why I chose this problem. The operator faces two costly mistakes. A missed leak can cause environmental harm and regulatory breaches that threaten its licence to operate, including possible revocation for serious or repeated violations. Conversely, mistaking a sensor fault for a leak can trigger an unnecessary offshore inspection involving personnel, vessels or an ROV. My PhD studied marine data quality, anomaly detection and the difficulty of translating expert knowledge into smart systems. I now want to apply that research in practice: make expert investigation knowledge reusable while delivering timely, evidence-backed warnings and inspection guidance.

Proposed workflow. Keep existing detectors. Agents observe the numerical evidence, choose read-only searches of incident and maintenance records, and select pressure-history comparisons to test competing explanations. They check document status and availability at the incident time, then present cited evidence, supported inspection areas, contradictions and uncertainty for human review. The LLM interprets the engineer's question and explains competing root-cause hypotheses in context: what supports them, what conflicts with them, and what the engineer should verify next. The review handoff combines this explanation with the checked inspection interval or explicit withholding, readable sources, measured comparisons and unresolved questions. Engineers can draft a follow-up against the same monitoring window and download a versioned evidence packet. Its citation-linked explanation is separate from checked measurements; the packet is not a confirmed RCA or engineer approval. A separate same-model critic reviews every explanation and verification question against original sources. One revision is allowed; rejected reports are withheld. Exact quotes and review coverage are code-checked, not a guarantee of correctness. A logged valve change may make an operating explanation plausible; it does not rule out a concurrent leak.

Why agents, not only ML? Numerical or graph-based methods can correlate measurements. The agent's additional role is interpreting engineer questions, deciding which historical evidence to investigate and explaining what it changes. A well-designed fixed workflow can also solve these small cases; additional model calls must earn their place against that cheaper baseline. A separate authored 16-case inspection-priority evaluation matched 15/16 priority sets with hosted interpretation versus 12/16 with context-aware rules. This tests interpreting operating records to prioritize numerically bracketed areas, not better leak coordinates or field accuracy. Deprioritized areas are not cleared of leaks.

Knowledge transfer. Versioned research-curated guidance and an investigation program are pinned with their provenance; incident records are synthetic. Agents apply the guidance to joint sensor comparisons and competing regional hypotheses, separately from numerical screening. This is not SME-approved or automatically acquired tacit expertise. In a customer pilot, experts must elicit, review and approve procedures and examples before reuse. Compare newcomer and experienced-engineer decisions; onboarding benefit is a hypothesis, not a measured result.

Customer questions. "Leak or faulty sensors?" "Where should we inspect, and why?" "What evidence supports this, and what remains unknown?"

Why unsigned. They doubt AI will reduce investigation effort. False alarms mean costly offshore inspections; missed leaks risk damage, disruption and regulatory action, including possible licence revocation for serious violations. Privacy, integration cost and accountability remain concerns.

Budget owner: Head of Subsea Operations. Cares about workload, inspection cost and safety. "What will this save, cost and take to deploy?"

Technical expert: Lead Subsea Integrity & Instrumentation Engineer. Cares about calibration, missing data and physical assumptions. "Can you distinguish pressure transients, shared failures and leaks, and reproduce the conclusion?"

Current implementation and limits. The prototype streams synthetic pressure readings through lightweight anomaly detectors and agent-assisted investigation, including gaps, spikes and drift. It recommends a station-supported inspection interval, not an exact leak position. A PostgreSQL full-text retrieval corpus contains fictional operating logs, maintenance reports and procedures, including superseded, expired and future-dated records. Searches and comparisons are scoped to the incident and retain executed tool receipts. The pressure profile is illustrative, not a validated hydraulic model. Customer detector, historian and maintenance-system integrations remain unimplemented; operating context and causation require human verification. No confidential data. Inference is hosted, not local-only. Read-only decision support, not certified protection or plant control; humans retain safety authority.

Pilot. Treat ten minutes per manual investigation as an assumption to validate. Aim for a usable AI report within five minutes; immediate numerical warnings are measured separately and never gated by inference. Compare accuracy, false/missed warnings, interval containment, unsupported claims, abstention and reliability under outages against manual/fixed baselines. Include failed and late attempts, not only successful answers. Zero observed unsupported claims is an acceptance goal, not a guarantee of hallucination-free language. Adopt only if evidence quality, knowledge transfer and total cost justify deployment.

Private demo repository: https://github.com/ntnguyen-so/subsea-integrity-copilot